Set 3C: Rectifiers and AC-Adapter Design

ECEN 325 — Lecture 3 (Diodes) practice set

What this set trains. Half-wave, center-tapped full-wave and bridge rectifiers: peak output, ripple with a filter capacitor, diode currents and peak inverse voltage — and, run backwards, the design of an AC adapter (transformer ratio, capacitor, diode ratings) from its specifications. All diodes use the constant-voltage-drop model (0.7 V); the mains is $120\,\mathrm{V_{rms}}$ (170 V peak), 60 Hz. Problems 1–6 are easy, 7–12 medium, 13–18 hard (design). Try each problem before looking at its solution.

The Lecture-3 pipeline (use it for every problem in this lecture)

Tools — this is all the theory you need (plus the Lecture-1/2 tools):

T1 *Diode models.* Ideal: OFF $\Leftrightarrow V_D < 0$ (open), ON $\Leftrightarrow I_D > 0$ ($V_D = 0$). Constant-voltage-drop (CVD): OFF $\Leftrightarrow V_D < 0.7\,\mathrm{V}$ (open), ON $\Leftrightarrow I_D > 0$ ($V_D = 0.7\,\mathrm{V}$ source). Exponential: $I_D = I_S e^{V_D/nV_T}$, i.e. $V_D = nV_T \ln(I_D/I_S)$, $V_T \approx 25\,\mathrm{mV}$.

T2 *State method.* Assume ON/OFF for every diode, solve the now-linear circuit (KCL, Lecture 2), then check the assumptions; each check is a condition on V_i. The valid states tile the V_i axis.

T3 *Small signal.* Around a DC point I_D the diode is a resistor $r_d = \frac{nV_T}{I_D}$ (valid while $|v_d| \ll nV_T$). Capacitors: open at DC; $\frac{1}{sC}$ in AC; DC supplies are AC ground.

T4 *Rectifier with a filter capacitor* ($f = 60\,\mathrm{Hz}$, $T = \frac{1}{f}$ half-wave, $\frac{1}{2f}$ full-wave): peak $V_p = \hat{V}_S -$ (diode drops in the path); ripple $V_r \approx V_p \frac{T}{RC}$; $I_{D,av} \approx I_L\left(1 + \pi\sqrt{2V_p/V_r}\right)$, $I_{D,max} \approx I_L\left(1 + 2\pi\sqrt{2V_p/V_r}\right)$; PIV = the largest reverse voltage a diode sees.

Steps:

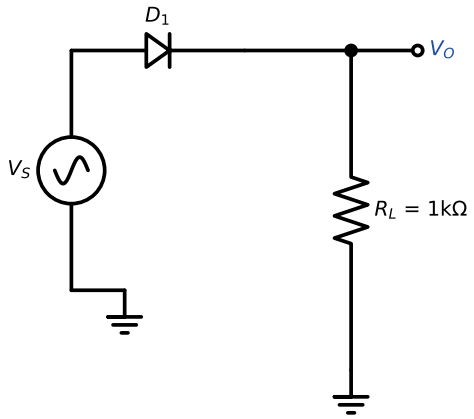
1. *Pick the model* stated in the problem.
2. *Transfer characteristic:* for each ON/OFF state: replace the diodes, solve, check $\Rightarrow$ range of V_i; then join the pieces (they must meet at the breakpoints) and sketch.
3. *Small signal:* DC operating point first, then r_d, then the AC circuit exactly as in Lectures 1–2, then add DC and AC.
4. *Rectifiers:* peak $\rightarrow$ ripple (or C for a given ripple) $\rightarrow$ diode currents $\rightarrow$ PIV $\rightarrow$ transformer ratio.

Problems

Easy problems

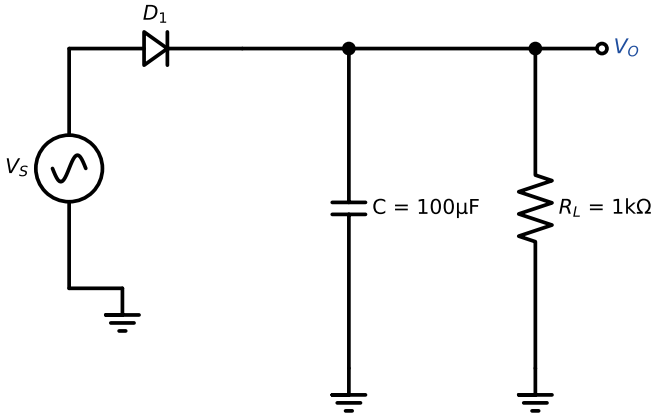
Problem 3C.1 — **Easy**

The half-wave rectifier below driven by a 60 Hz sinusoid of peak $\hat{V}_S = 10\,\text{V}$ drives $R_L = 1\,\text{k}\Omega$ (no filter capacitor). Using the constant-voltage-drop model (0.7 V), find the peak output voltage and the peak inverse voltage (PIV) of the diodes, and sketch $V_O(t)$.



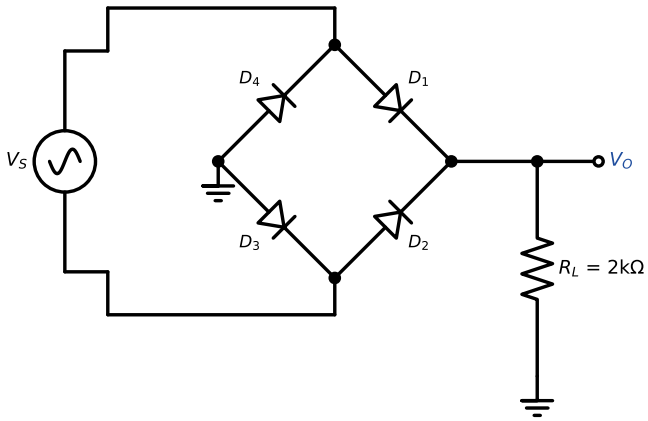
Problem 3C.2 — **Easy**

The half-wave rectifier below driven by a 60 Hz sinusoid of peak $\hat{V}_S = 12\,\text{V}$ drives $R_L = 1\,\text{k}\Omega$ with a filter capacitor $C = 100\,\mu\text{F}$. Using the constant-voltage-drop model (0.7 V), find the peak output V_p, the ripple V_r, the minimum output, the diode currents $I_{D,av}$ and $I_{D,max}$, and the PIV.



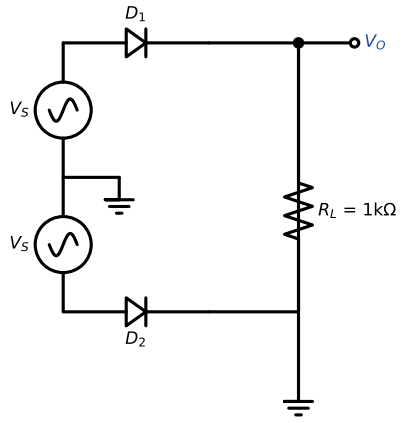
Problem 3C.3 — Easy

The bridge rectifier below driven by a 60 Hz sinusoid of peak $\hat{V}_S = 15\,\text{V}$ drives $R_L = 2\,\text{k}\Omega$ (no filter capacitor). Using the constant-voltage-drop model (0.7 V), find the peak output voltage and the peak inverse voltage (PIV) of the diodes, and sketch $V_O(t)$.



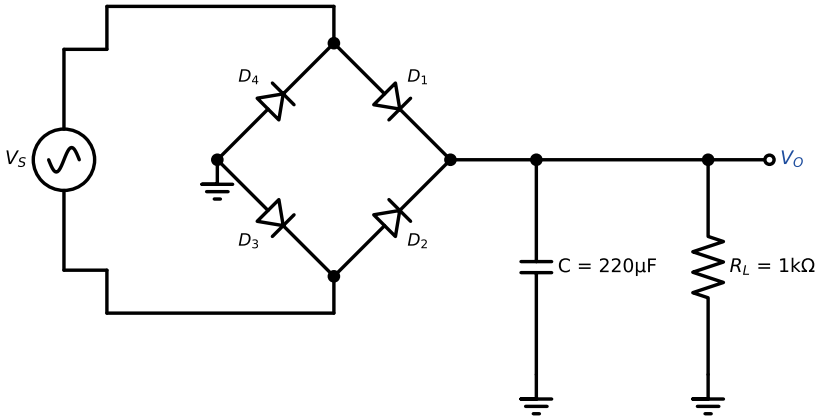
Problem 3C.4 — Easy

The center-tapped full-wave rectifier below driven by a 60 Hz sinusoid of peak $\hat{V}_S = 9\,\text{V}$ on each half of the secondary drives $R_L = 1\,\text{k}\Omega$ (no filter capacitor). Using the constant-voltage-drop model (0.7 V), find the peak output voltage and the peak inverse voltage (PIV) of the diodes, and sketch $V_O(t)$.



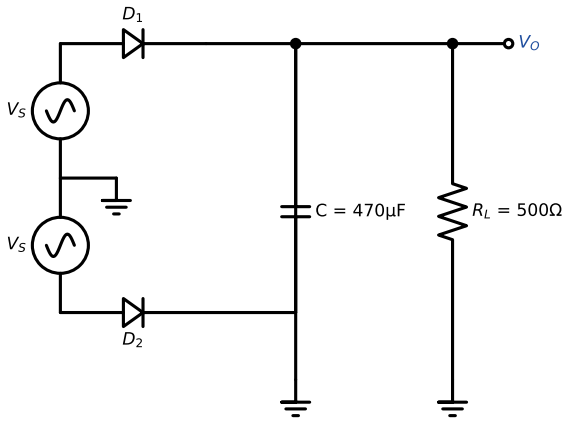
Problem 3C.5 — Easy

The bridge rectifier below driven by a 60 Hz sinusoid of peak $\hat{V}_S = 12\,\text{V}$ drives $R_L = 1\,\text{k}\Omega$ with a filter capacitor $C = 220\,\mu\text{F}$. Using the constant-voltage-drop model (0.7 V), find the peak output V_p, the ripple V_r, the minimum output, the diode currents $I_{D,av}$ and $I_{D,max}$, and the PIV.



Problem 3C.6 — Easy

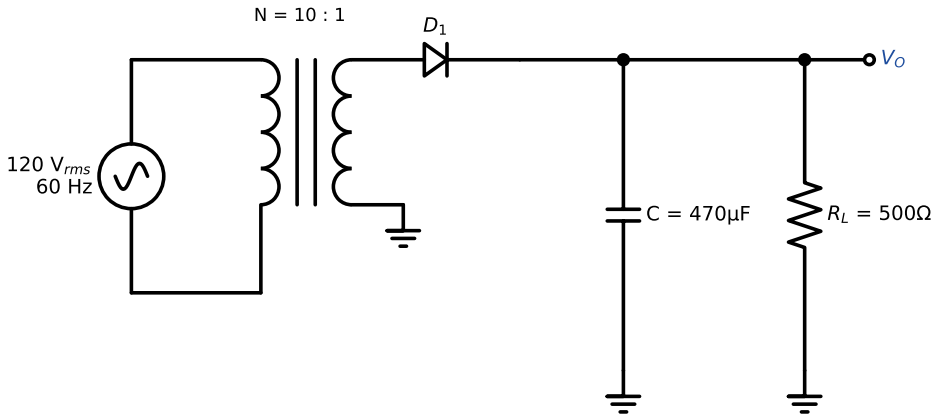
The center-tapped full-wave rectifier below driven by a 60 Hz sinusoid of peak $\hat{V}_S = 10\,\text{V}$ on each half of the secondary drives $R_L = 500\,\Omega$ with a filter capacitor $C = 470\,\mu\text{F}$. Using the constant-voltage-drop model (0.7 V), find the peak output V_p, the ripple V_r, the minimum output, the diode currents $I_{D,av}$ and $I_{D,max}$, and the PIV.



Medium problems

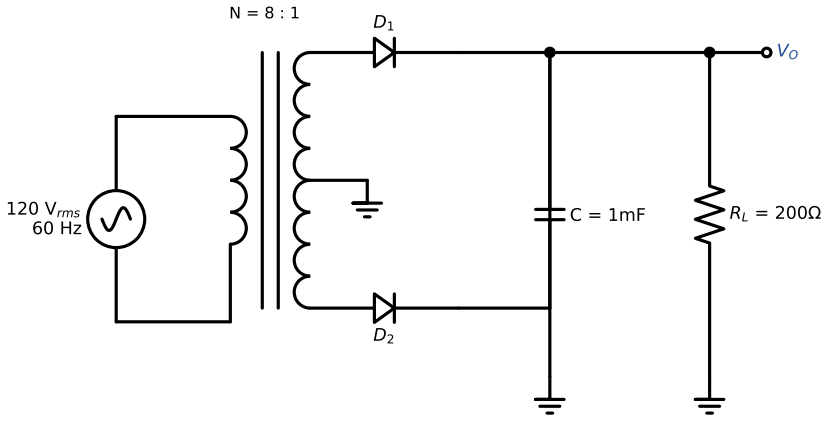
Problem 3C.7 — Medium

The half-wave rectifier below fed from the $120\,\mathrm{V_{rms}}$, $60\,\mathrm{Hz}$ mains through a transformer with turns ratio $N = 10:1$ drives $R_L = 500\,\Omega$ with a filter capacitor $C = 470\,\mu\mathrm{F}$. Using the constant-voltage-drop model ($0.7\,\mathrm{V}$), find the peak output V_p, the ripple V_r, the minimum output, the diode currents $I_{D,av}$ and $I_{D,max}$, and the PIV.



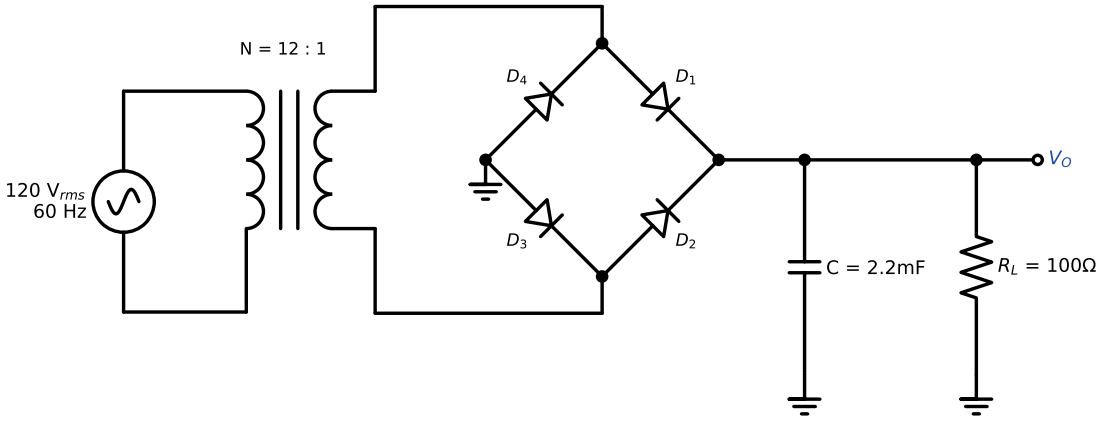
Problem 3C.8 — Medium

The center-tapped full-wave rectifier below fed from the $120\,\mathrm{V_{rms}}$, $60\,\mathrm{Hz}$ mains through a transformer with turns ratio $N = 8:1$ (primary : whole secondary) drives $R_L = 200\,\Omega$ with a filter capacitor $C = 1\,\mathrm{mF}$. Using the constant-voltage-drop model ($0.7\,\mathrm{V}$), find the peak output V_p, the ripple V_r, the minimum output, the diode currents $I_{D,av}$ and $I_{D,max}$, and the PIV.



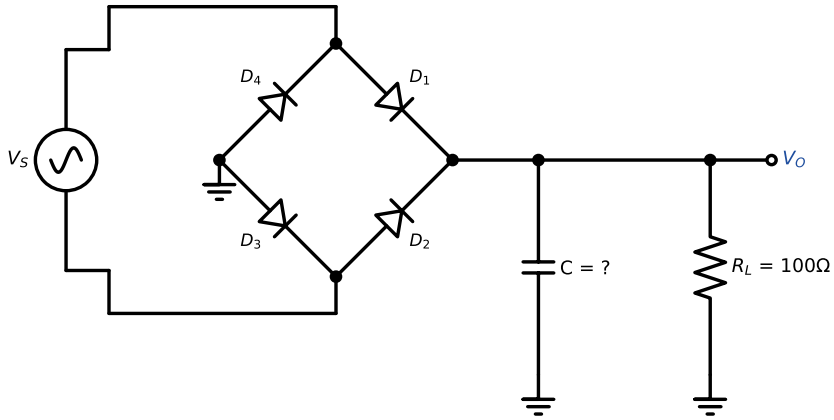
Problem 3C.9 — Medium

The bridge rectifier below fed from the $120\,\mathrm{V_{rms}}$, $60\,\mathrm{Hz}$ mains through a transformer with turns ratio $N = 12:1$ drives $R_L = 100\,\Omega$ with a filter capacitor $C = 2.2\,\mathrm{mF}$. Using the constant-voltage-drop model ($0.7\,\mathrm{V}$), find the peak output V_p, the ripple V_r, the minimum output, the diode currents $I_{D,av}$ and $I_{D,max}$, and the PIV.



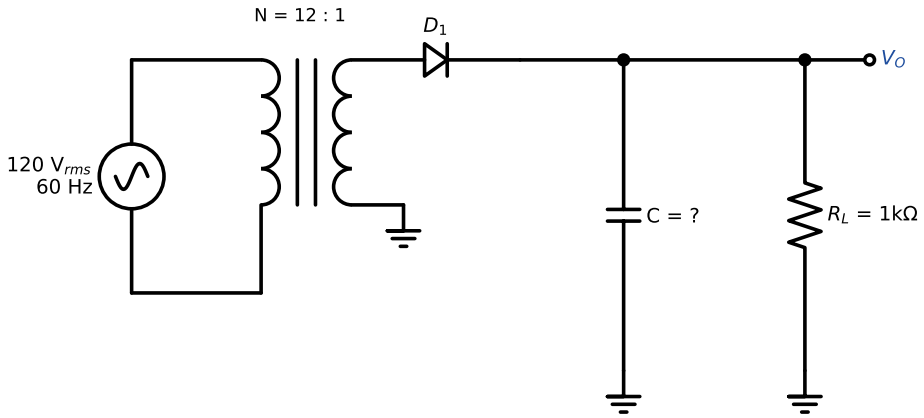
Problem 3C.10 — Medium

The bridge rectifier below driven by a $60\,\mathrm{Hz}$ sinusoid of peak $\hat{V}_S = 10\,\mathrm{V}$ drives $R_L = 100\,\Omega$. Find the filter capacitance C for a peak-to-peak ripple of at most $500\,\mathrm{mV}$, the diode currents $I_{D,av}$, $I_{D,max}$ and the peak inverse voltage (PIV) of the diodes. Use the constant-voltage-drop model ($0.7\,\mathrm{V}$).



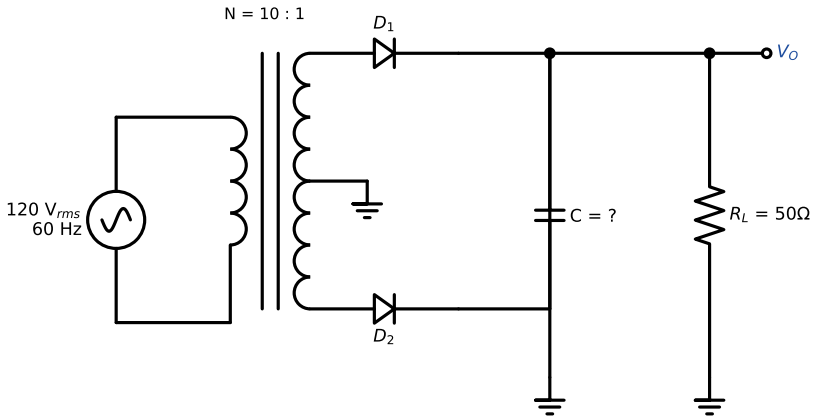
Problem 3C.11 — Medium

The half-wave rectifier below fed from the $120\,\mathrm{V_{rms}}$, $60\,\mathrm{Hz}$ mains through a transformer with turns ratio $N = 12:1$ drives $R_L = 1\,\mathrm{k\Omega}$. Find the filter capacitance C for an output that never drops below $11\,\mathrm{V}$, the diode currents $I_{D,av}$, $I_{D,max}$ and the peak inverse voltage (PIV) of the diodes. Use the constant-voltage-drop model ($0.7\,\mathrm{V}$).



Problem 3C.12 — Medium

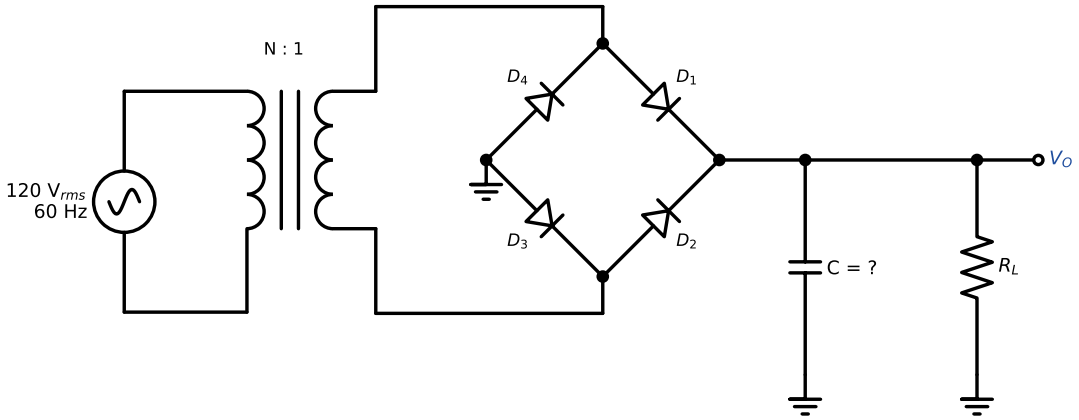
The center-tapped full-wave rectifier below fed from the $120\,\mathrm{V_{rms}}$, $60\,\mathrm{Hz}$ mains through a transformer with turns ratio $N = 10:1$ (primary : whole secondary) drives $R_L = 50\,\Omega$. Find the filter capacitance C for a peak-to-peak ripple of at most $400\,\mathrm{mV}$, the diode currents $I_{D,av}$, $I_{D,max}$ and the peak inverse voltage (PIV) of the diodes. Use the constant-voltage-drop model ($0.7\,\mathrm{V}$).



Hard problems

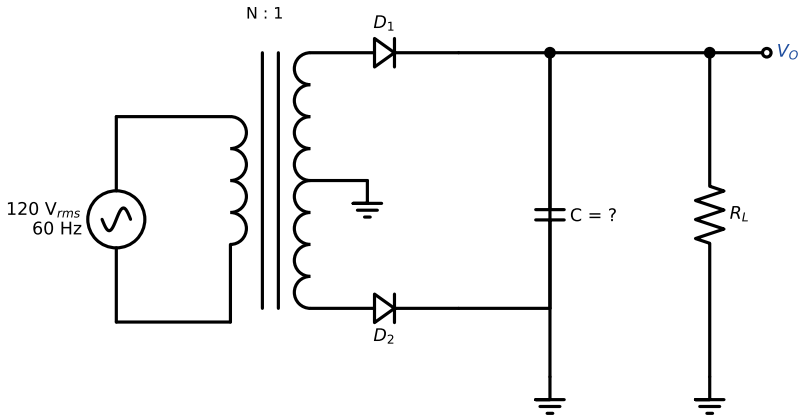
Problem 3C.13 — Hard

Design an AC adapter using a single transformer and a bridge rectifier (constant-voltage-drop diodes, 0.7 V; mains 120 V_{rms}, 60 Hz) for: output voltage $V_o = 3$ V, maximum ripple 5%, load current 0–400 mA. Find the transformer ratio N, the capacitance C and the minimum diode ratings ($I_{D,max}$, $I_{D,av}$, reverse breakdown voltage).



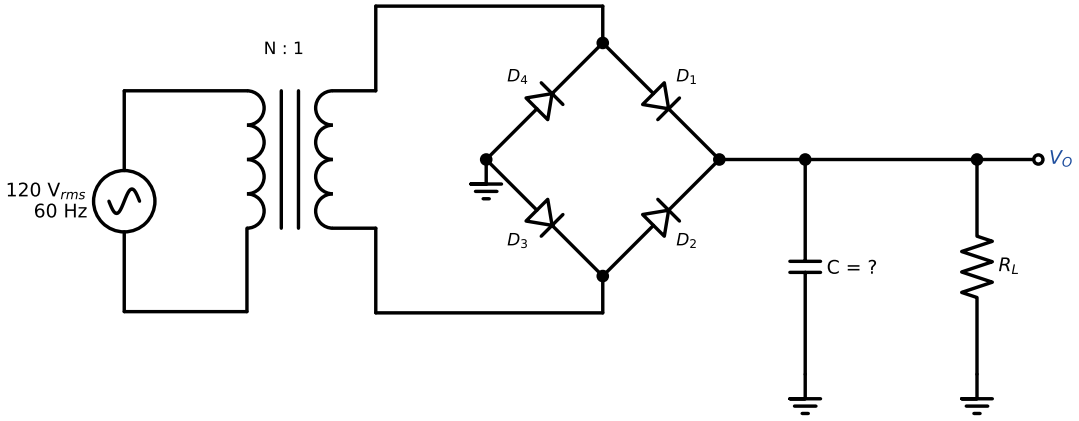
Problem 3C.14 — Hard

Design an AC adapter using a center-tapped transformer and a full-wave rectifier (constant-voltage-drop diodes, 0.7 V; mains 120 V_{rms}, 60 Hz) for: output voltage $V_o = 5$ V, maximum ripple 10%, load current 0–1.5 A. Find the transformer ratio N, the capacitance C and the minimum diode ratings ($I_{D,max}$, $I_{D,av}$, reverse breakdown voltage).



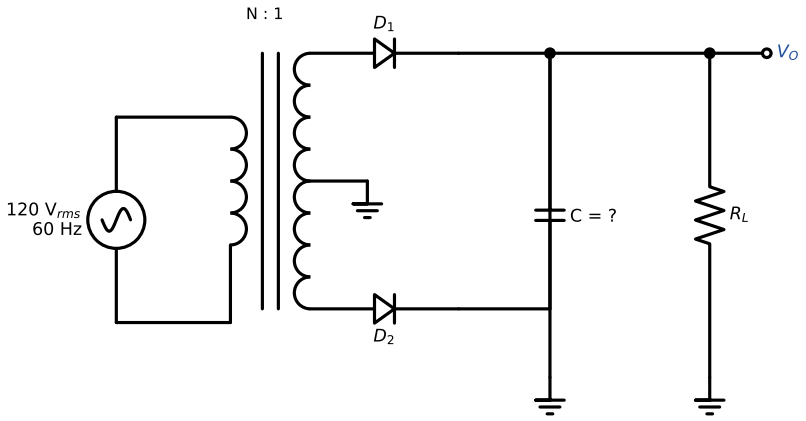
Problem 3C.15 — Hard

Design an AC adapter using a single transformer and a bridge rectifier (constant-voltage-drop diodes, 0.7 V; mains 120 V_{rms}, 60 Hz) for: output voltage $V_o = 12\,\text{V}$, maximum ripple 5%, load current 0–1 A. Find the transformer ratio N, the capacitance C and the minimum diode ratings ($I_{D,max}$, $I_{D,av}$, reverse breakdown voltage).



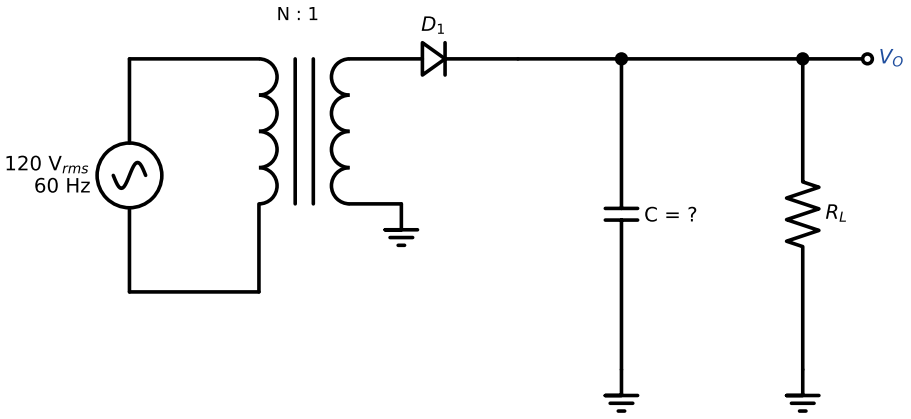
Problem 3C.16 — Hard

Design an AC adapter using a center-tapped transformer and a full-wave rectifier (constant-voltage-drop diodes, 0.7 V; mains 120 V_{rms}, 60 Hz) for: output voltage $V_o = 9\,\text{V}$, maximum ripple 5%, load current 0–500 mA. Find the transformer ratio N, the capacitance C and the minimum diode ratings ($I_{D,max}$, $I_{D,av}$, reverse breakdown voltage).



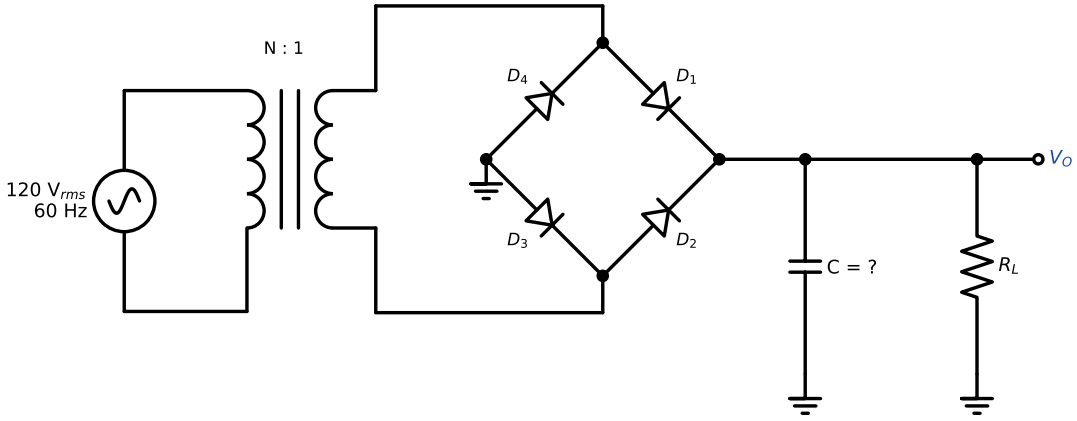
Problem 3C.17 — Hard

Design an AC adapter using a transformer and a half-wave rectifier (constant-voltage-drop diodes, 0.7 V; mains 120 V_{rms}, 60 Hz) for: output voltage $V_o = 5$ V, maximum ripple 10%, load current 0–100 mA. Find the transformer ratio N, the capacitance C and the minimum diode ratings ($I_{D,max}$, $I_{D,av}$, reverse breakdown voltage).



Problem 3C.18 — Hard

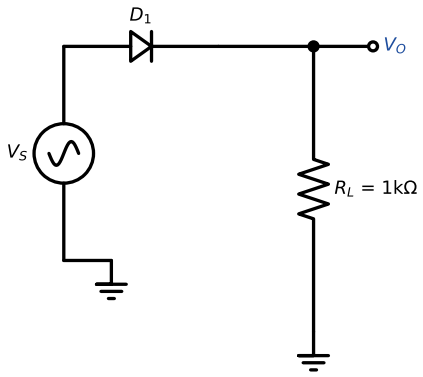
Design an AC adapter using a single transformer and a bridge rectifier (constant-voltage-drop diodes, 0.7 V; mains 120 V_{rms}, 60 Hz) for: output voltage $V_o = 15$ V, maximum ripple 5%, load current 0–800 mA. Find the transformer ratio N, the capacitance C and the minimum diode ratings ($I_{D,max}$, $I_{D,av}$, reverse breakdown voltage).



Solutions

Every solution follows the steps of the Lecture-3 pipeline card.

Solution 3C.1 **Easy**



Step 1 — Peak output

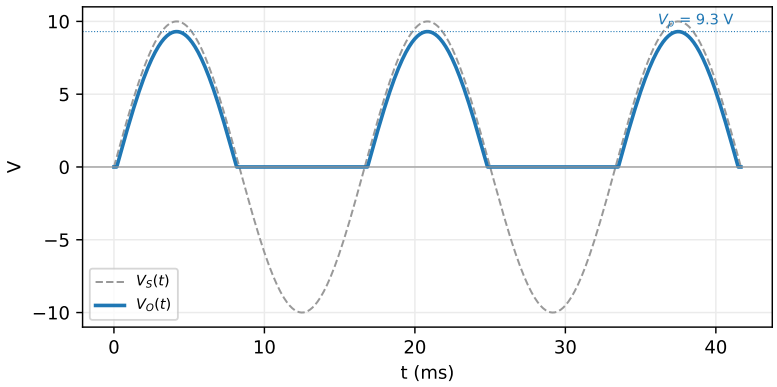
At the peak of the input the current flows through the single diode, each dropping 0.7 V (tool T1):

$$V_p = \hat{V}_S - 0.7 = 10 - 0.7 = 9.3\,\text{V}, \qquad I_L \approx \frac{V_p}{R_L} = 9.3\,\text{mA}$$

Step 2 — Peak inverse voltage

At the negative peak the diode is OFF, the output is 0 and the whole $\hat{V}_S$ appears across the diode: PIV $= \hat{V}_S = 10\,\text{V}$. The diodes must have a reverse breakdown voltage above the PIV.

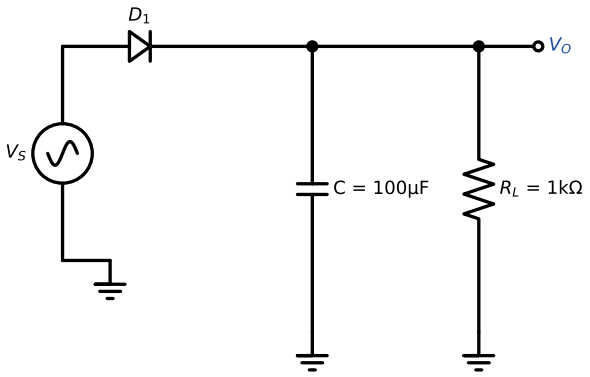
Step 3 — Waveforms



Answer

$V_p = 9.3\,\text{V}, \quad \text{PIV} = 10\,\text{V}$

Solution 3C.2 **Easy**



Step 1 — Peak output

At the peak of the input the current flows through the single diode, each dropping 0.7 V (tool T1):

$$V_p = \hat{V}_S - 0.7 = 12 - 0.7 = 11.3\,\text{V}, \qquad I_L \approx \frac{V_p}{R_L} = 11.3\,\text{mA}$$

Step 2 — Ripple

Half-wave: the capacitor is recharged once per period, $T = 1/60\,\text{s}$. Between recharges the capacitor discharges through R_L (tool T4):

$$V_r \approx V_p \frac{T}{R_L C} = \frac{11.3 \times 0.01667}{1000 \times 10^{-4}} = 1.88\,\text{V}$$

so the output swings between $V_p = 11.3\,\text{V}$ and $V_p - V_r = 9.42\,\text{V}$ (average $\approx V_p - V_r/2 = 10.4\,\text{V}$).

Step 3 — Diode currents

The diodes only conduct during the short recharge interval, so they must deliver a whole period's charge in a short burst (tool T4):

$$\sqrt{2V_p/V_r} = \sqrt{2 \times 11.3/1.883} = 3.464$$

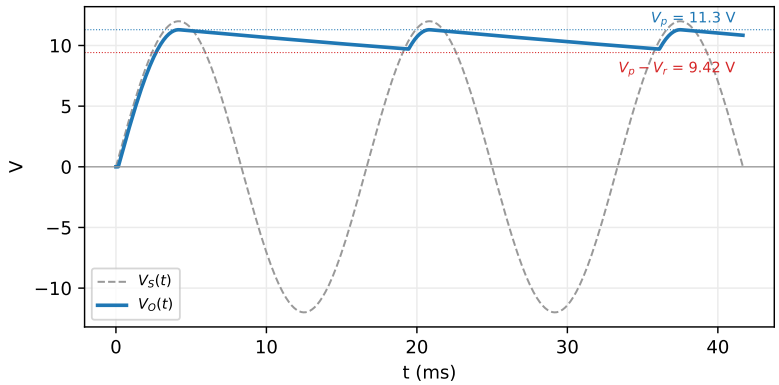
$$I_{D,av} = I_L\left(1 + \pi\sqrt{2V_p/V_r}\right) = 0.0113\,(1 + \pi \times 3.464) = 134\,\text{mA}$$

$$I_{D,max} = I_L\left(1 + 2\pi\sqrt{2V_p/V_r}\right) = 257\,\text{mA}$$

Step 4 — Peak inverse voltage

At the negative peak the anode is at $-\hat{V}_S$ while the capacitor holds the cathode near V_p: PIV $= \hat{V}_S + V_p = 2\hat{V}_S - 0.7 = 23.3\,\text{V}$. The diodes must have a reverse breakdown voltage above the PIV.

Step 5 — Waveforms

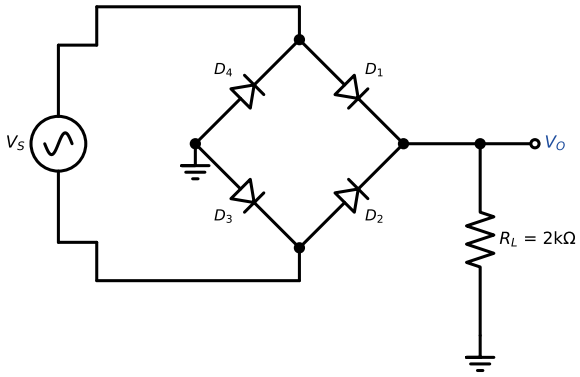


The ripple formula assumes the capacitor discharges for a whole T; in reality it is recharged a little earlier, so the true ripple (plotted) is slightly smaller — the formula errs on the safe side.

Answer

$V_p = 11.3\,\text{V}$, $V_r = 1.88\,\text{V}$, $V_{o,min} = 9.42\,\text{V}$, $I_{D,av} = 134\,\text{mA}$, $I_{D,max} = 257\,\text{mA}$, PIV $= 23.3\,\text{V}$

Solution 3C.3 **Easy**



Step 1 — Peak output

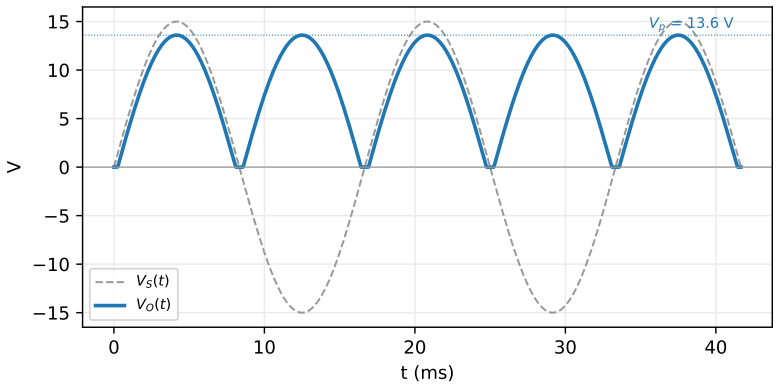
At the peak of the input the current flows through two diodes in series (one on each side of the bridge), each dropping 0.7 V (tool T1):

$$V_p = \hat{V}_S - 1.4 = 15 - 1.4 = 13.6\,\text{V}, \qquad I_L \approx \frac{V_p}{R_L} = 6.8\,\text{mA}$$

Step 2 — Peak inverse voltage

When a pair of diodes conducts, each of the other two sees the output plus one diode drop: PIV $= V_p + 0.7 = \hat{V}_S - 0.7 = 14.3\,\text{V}$. The diodes must have a reverse breakdown voltage above the PIV.

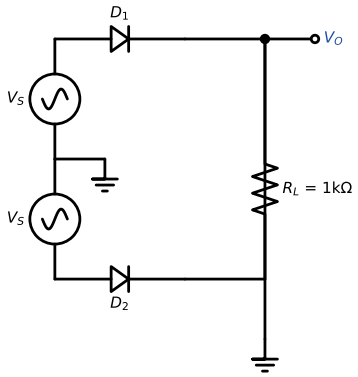
Step 3 — Waveforms



Answer

$V_p = 13.6\,\text{V}, \quad \text{PIV} = 14.3\,\text{V}$

Solution 3C.4 **Easy**



Step 1 — Peak output

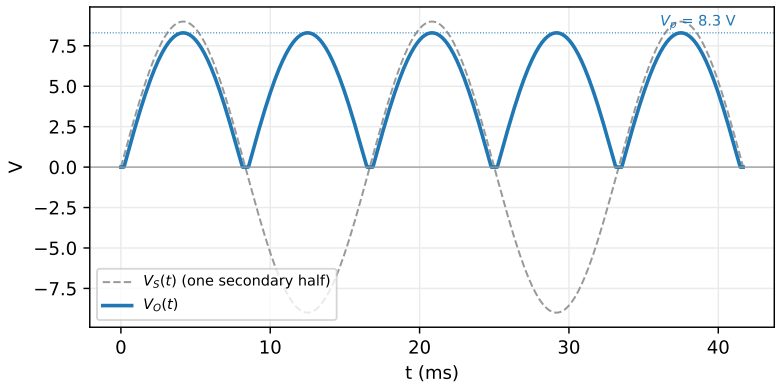
At the peak of the input the current flows through one diode (each half of the secondary has its own diode), each dropping 0.7 V (tool T1):

$$V_p = \hat{V}_S - 0.7 = 9 - 0.7 = 8.3\,\text{V}, \qquad I_L \approx \frac{V_p}{R_L} = 8.3\,\text{mA}$$

Step 2 — Peak inverse voltage

When one diode conducts its cathode sits at $V_p = \hat{V}_S - 0.7$ while its own half of the secondary is at $-\hat{V}_S$: $\text{PIV} = 2\hat{V}_S - 0.7 = 17.3\,\text{V}$. The diodes must have a reverse breakdown voltage above the PIV.

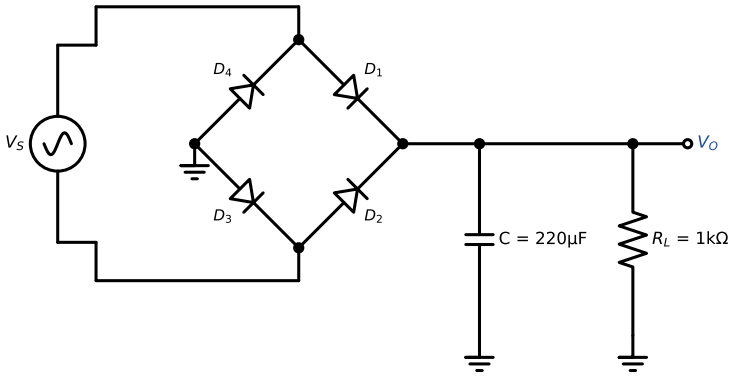
Step 3 — Waveforms



Answer

$V_p = 8.3\,\text{V}, \quad \text{PIV} = 17.3\,\text{V}$

Solution 3C.5 **Easy**



Step 1 — Peak output

At the peak of the input the current flows through two diodes in series (one on each side of the bridge), each dropping 0.7 V (tool T1):

$$V_p = \hat{V}_S - 1.4 = 12 - 1.4 = 10.6\,\text{V}, \qquad I_L \approx \frac{V_p}{R_L} = 10.6\,\text{mA}$$

Step 2 — Ripple

Full-wave: the capacitor is recharged every half period, $T = 1/120\,\text{s}$. Between recharges the capacitor discharges through R_L (tool T4):

$$V_r \approx V_p \frac{T}{R_L C} = \frac{10.6 \times 0.008333}{1000 \times 2.2 \times 10^{-4}} = 402\,\text{mV}$$

so the output swings between $V_p = 10.6\,\text{V}$ and $V_p - V_r = 10.2\,\text{V}$ (average $\approx V_p - V_r/2 = 10.4\,\text{V}$).

Step 3 — Diode currents

The diodes only conduct during the short recharge interval, so they must deliver a whole period's charge in a short burst (tool T4):

$$\sqrt{2V_p/V_r} = \sqrt{2 \times 10.6/0.4015} = 7.266$$

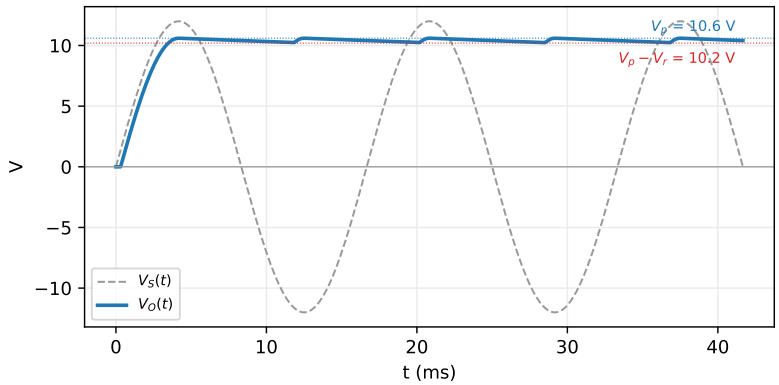
$$I_{D,av} = I_L\left(1 + \pi\sqrt{2V_p/V_r}\right) = 0.0106\,(1 + \pi \times 7.266) = 253\,\text{mA}$$

$$I_{D,max} = I_L\left(1 + 2\pi\sqrt{2V_p/V_r}\right) = 495\,\text{mA}$$

Step 4 — Peak inverse voltage

When a pair of diodes conducts, each of the other two sees the output plus one diode drop: PIV $= V_p + 0.7 = \hat{V}_S - 0.7 = 11.3\,\text{V}$. The diodes must have a reverse breakdown voltage above the PIV.

Step 5 — Waveforms

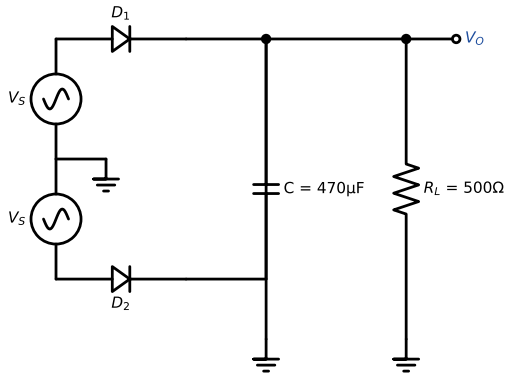


The ripple formula assumes the capacitor discharges for a whole T; in reality it is recharged a little earlier, so the true ripple (plotted) is slightly smaller — the formula errs on the safe side.

Answer

$V_p = 10.6\,\text{V}$, $V_r = 402\,\text{mV}$, $V_{o,min} = 10.2\,\text{V}$, $I_{D,av} = 253\,\text{mA}$, $I_{D,max} = 495\,\text{mA}$, PIV $= 11.3\,\text{V}$

Solution 3C.6 **Easy**



Step 1 — Peak output

At the peak of the input the current flows through one diode (each half of the secondary has its own diode), each dropping 0.7 V (tool T1):

$$V_p = \hat{V}_S - 0.7 = 10 - 0.7 = 9.3\,\text{V}, \qquad I_L \approx \frac{V_p}{R_L} = 18.6\,\text{mA}$$

Step 2 — Ripple

Full-wave: the capacitor is recharged every half period, $T = 1/120\,\text{s}$. Between recharges the capacitor discharges through R_L (tool T4):

$$V_r \approx V_p \frac{T}{R_L C} = \frac{9.3 \times 0.008333}{500 \times 4.7 \times 10^{-4}} = 330\,\text{mV}$$

so the output swings between $V_p = 9.3\,\text{V}$ and $V_p - V_r = 8.97\,\text{V}$ (average $\approx V_p - V_r/2 = 9.14\,\text{V}$).

Step 3 — Diode currents

The diodes only conduct during the short recharge interval, so they must deliver a whole period's charge in a short burst (tool T4):

$$\sqrt{2V_p/V_r} = \sqrt{2 \times 9.3/0.3298} = 7.51$$

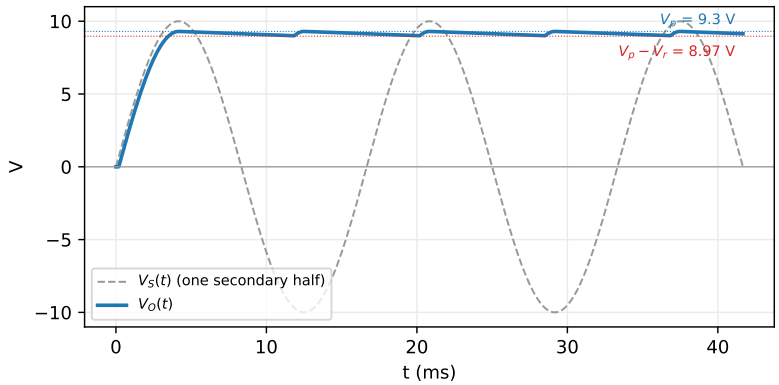
$$I_{D,av} = I_L\left(1 + \pi\sqrt{2V_p/V_r}\right) = 0.0186\,(1 + \pi \times 7.51) = 457\,\text{mA}$$

$$I_{D,max} = I_L\left(1 + 2\pi\sqrt{2V_p/V_r}\right) = 896\,\text{mA}$$

Step 4 — Peak inverse voltage

When one diode conducts its cathode sits at $V_p = \hat{V}_S - 0.7$ while its own half of the secondary is at $-\hat{V}_S$: $\text{PIV} = 2\hat{V}_S - 0.7 = 19.3\,\text{V}$. The diodes must have a reverse breakdown voltage above the PIV.

Step 5 — Waveforms

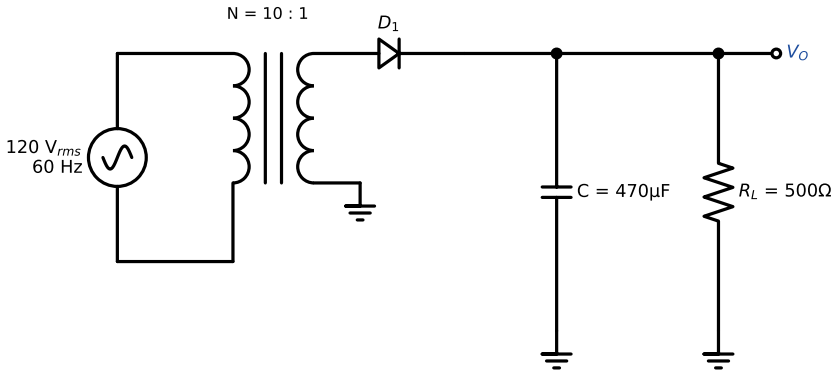


The ripple formula assumes the capacitor discharges for a whole T; in reality it is recharged a little earlier, so the true ripple (plotted) is slightly smaller — the formula errs on the safe side.

Answer

$V_p = 9.3\,\text{V}$, $V_r = 330\,\text{mV}$, $V_{o,min} = 8.97\,\text{V}$, $I_{D,av} = 457\,\text{mA}$, $I_{D,max} = 896\,\text{mA}$, PIV $= 19.3\,\text{V}$

Solution 3C.7 **Medium**



Step 1 — Secondary voltage

The mains peak is $120\sqrt{2} = 170\,\text{V}$, so the secondary peaks at $\frac{170}{N} = 17\,\text{V} = \hat{V}_S$.

Step 2 — Peak output

At the peak of the input the current flows through the single diode, each dropping 0.7 V (tool T1):

$$V_p = \hat{V}_S - 0.7 = 17 - 0.7 = 16.3\,\text{V}, \qquad I_L \approx \frac{V_p}{R_L} = 32.6\,\text{mA}$$

Step 3 — Ripple

Half-wave: the capacitor is recharged once per period, $T = 1/60\,\text{s}$. Between recharges the capacitor discharges through R_L (tool T4):

$$V_r \approx V_p \frac{T}{R_L C} = \frac{16.3 \times 0.01667}{500 \times 4.7 \times 10^{-4}} = 1.16\,\text{V}$$

so the output swings between $V_p = 16.3\,\text{V}$ and $V_p - V_r = 15.1\,\text{V}$ (average $\approx V_p - V_r/2 = 15.7\,\text{V}$).

Step 4 — Diode currents

The diodes only conduct during the short recharge interval, so they must deliver a whole period's charge in a short burst (tool T4):

$$\sqrt{2V_p/V_r} = \sqrt{2 \times 16.3/1.156} = 5.31$$

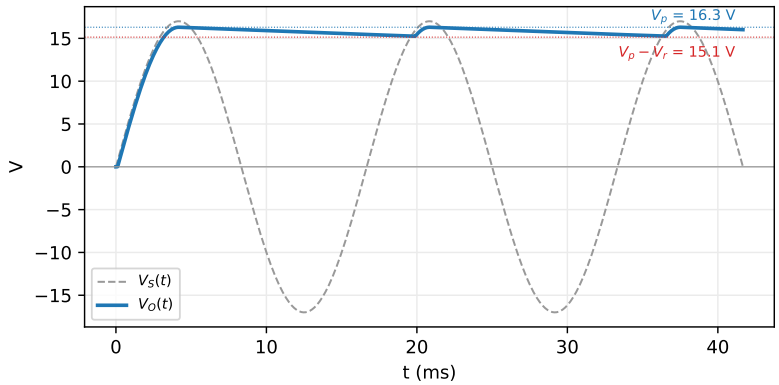
$$I_{D,av} = I_L\left(1 + \pi\sqrt{2V_p/V_r}\right) = 0.0326\,(1 + \pi \times 5.31) = 576\,\text{mA}$$

$$I_{D,max} = I_L\left(1 + 2\pi\sqrt{2V_p/V_r}\right) = 1.12\,\text{A}$$

Step 5 — Peak inverse voltage

At the negative peak the anode is at $-\hat{V}_S$ while the capacitor holds the cathode near V_p: PIV $= \hat{V}_S + V_p = 2\hat{V}_S - 0.7 = 33.3\,\text{V}$. The diodes must have a reverse breakdown voltage above the PIV.

Step 6 — Waveforms

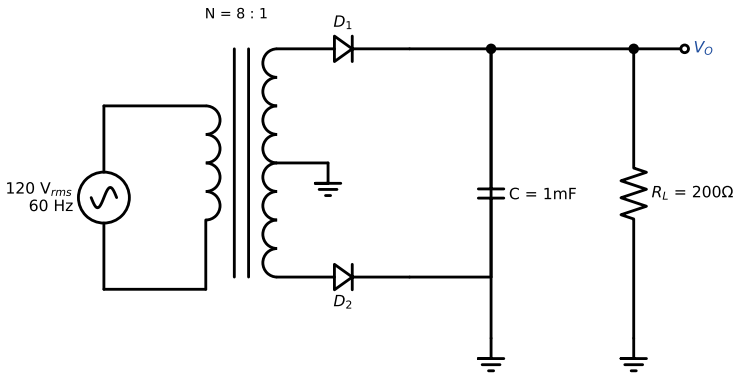


The ripple formula assumes the capacitor discharges for a whole T; in reality it is recharged a little earlier, so the true ripple (plotted) is slightly smaller — the formula errs on the safe side.

Answer

$V_p = 16.3\,\text{V}$, $V_r = 1.16\,\text{V}$, $V_{o,min} = 15.1\,\text{V}$, $I_{D,av} = 576\,\text{mA}$, $I_{D,max} = 1.12\,\text{A}$, $\text{PIV} = 33.3\,\text{V}$

Solution 3C.8 **Medium**



Step 1 — Secondary voltage

The mains peak is $120\sqrt{2} = 170\,\text{V}$, so the secondary peaks at $\frac{170}{N} = 21.2\,\text{V}$; each half of the center-tapped secondary gets $\hat{V}_S = 10.6\,\text{V}$.

Step 2 — Peak output

At the peak of the input the current flows through one diode (each half of the secondary has its own diode), each dropping 0.7 V (tool T1):

$$V_p = \hat{V}_S - 0.7 = 10.62 - 0.7 = 9.93\,\text{V}, \qquad I_L \approx \frac{V_p}{R_L} = 49.6\,\text{mA}$$

Step 3 — Ripple

Full-wave: the capacitor is recharged every half period, $T = 1/120\,\text{s}$. Between recharges the capacitor discharges through R_L (tool T4):

$$V_r \approx V_p \frac{T}{R_L C} = \frac{9.925 \times 0.008333}{200 \times 0.001} = 414\,\text{mV}$$

so the output swings between $V_p = 9.93\,\text{V}$ and $V_p - V_r = 9.51\,\text{V}$ (average $\approx V_p - V_r/2 = 9.72\,\text{V}$).

Step 4 — Diode currents

The diodes only conduct during the short recharge interval, so they must deliver a whole period's charge in a short burst (tool T4):

$$\sqrt{2V_p/V_r} = \sqrt{2 \times 9.925/0.4135} = 6.928$$

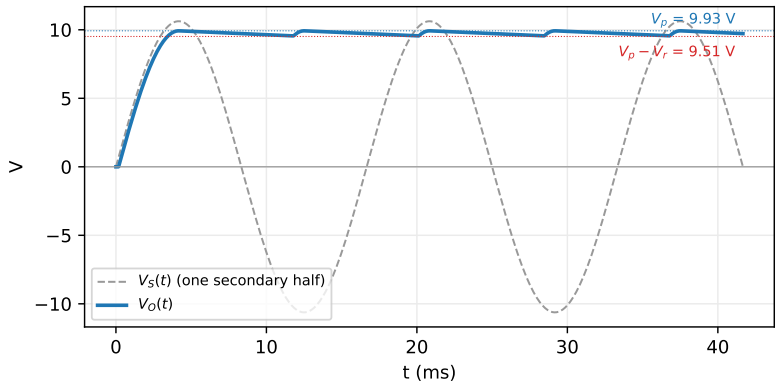
$$I_{D,av} = I_L\left(1 + \pi\sqrt{2V_p/V_r}\right) = 0.04963\,(1 + \pi \times 6.928) = 1.13\,\text{A}$$

$$I_{D,max} = I_L\left(1 + 2\pi\sqrt{2V_p/V_r}\right) = 2.21\,\text{A}$$

Step 5 — Peak inverse voltage

When one diode conducts its cathode sits at $V_p = \hat{V}_S - 0.7$ while its own half of the secondary is at $-\hat{V}_S$: $\text{PIV} = 2\hat{V}_S - 0.7 = 20.6\,\text{V}$. The diodes must have a reverse breakdown voltage above the PIV.

Step 6 — Waveforms

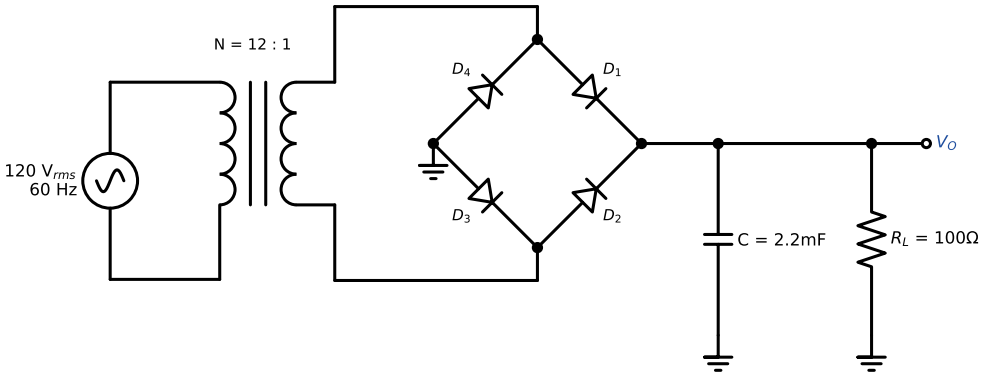


The ripple formula assumes the capacitor discharges for a whole T; in reality it is recharged a little earlier, so the true ripple (plotted) is slightly smaller — the formula errs on the safe side.

Answer

$V_p = 9.93\,\text{V}, \quad V_r = 414\,\text{mV}, \quad V_{o,min} = 9.51\,\text{V}, \quad I_{D,av} = 1.13\,\text{A}, \quad I_{D,max} = 2.21\,\text{A}, \quad \text{PIV} = 20.6\,\text{V}$

Solution 3C.9 **Medium**



Step 1 — Secondary voltage

The mains peak is $120\sqrt{2} = 170\,\text{V}$, so the secondary peaks at $\frac{170}{N} = 14.2\,\text{V} = \hat{V}_S$.

Step 2 — Peak output

At the peak of the input the current flows through two diodes in series (one on each side of the bridge), each dropping 0.7 V (tool T1):

$$V_p = \hat{V}_S - 1.4 = 14.17 - 1.4 = 12.8\,\text{V}, \qquad I_L \approx \frac{V_p}{R_L} = 128\,\text{mA}$$

Step 3 — Ripple

Full-wave: the capacitor is recharged every half period, $T = 1/120\,\text{s}$. Between recharges the capacitor discharges through R_L (tool T4):

$$V_r \approx V_p \frac{T}{R_L C} = \frac{12.77 \times 0.008333}{100 \times 0.0022} = 484\,\text{mV}$$

so the output swings between $V_p = 12.8\,\text{V}$ and $V_p - V_r = 12.3\,\text{V}$ (average $\approx V_p - V_r/2 = 12.5\,\text{V}$).

Step 4 — Diode currents

The diodes only conduct during the short recharge interval, so they must deliver a whole period's charge in a short burst (tool T4):

$$\sqrt{2V_p/V_r} = \sqrt{2 \times 12.77/0.4836} = 7.266$$

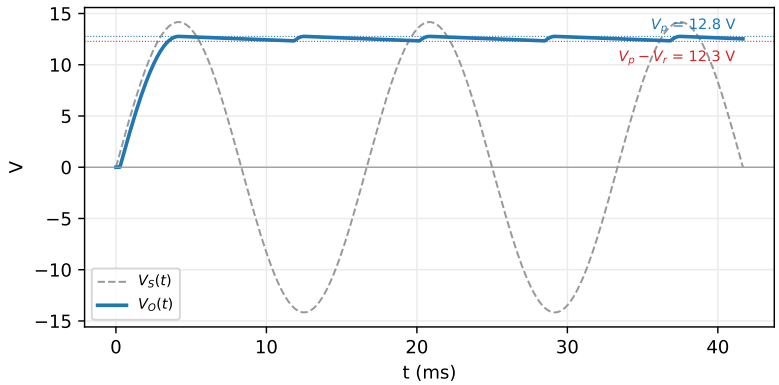
$$I_{D,av} = I_L\left(1 + \pi\sqrt{2V_p/V_r}\right) = 0.1277\,(1 + \pi \times 7.266) = 3.04\,\text{A}$$

$$I_{D,max} = I_L\left(1 + 2\pi\sqrt{2V_p/V_r}\right) = 5.96\,\text{A}$$

Step 5 — Peak inverse voltage

When a pair of diodes conducts, each of the other two sees the output plus one diode drop: PIV $= V_p + 0.7 = \hat{V}_S - 0.7 = 13.5\,\text{V}$. The diodes must have a reverse breakdown voltage above the PIV.

Step 6 — Waveforms

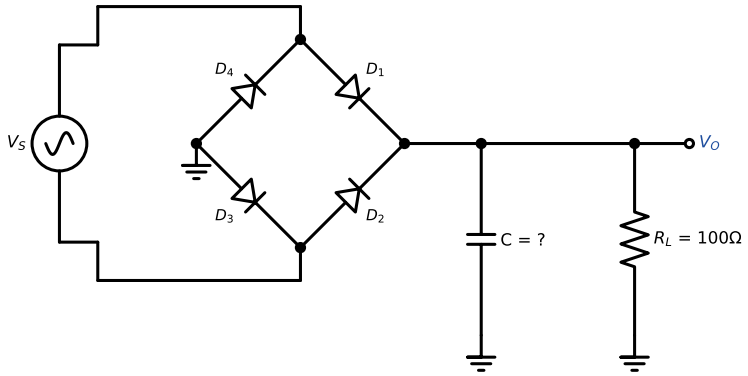


The ripple formula assumes the capacitor discharges for a whole T; in reality it is recharged a little earlier, so the true ripple (plotted) is slightly smaller — the formula errs on the safe side.

Answer

$V_p = 12.8\,\mathrm{V}$, $V_r = 484\,\mathrm{mV}$, $V_{o,min} = 12.3\,\mathrm{V}$, $I_{D,av} = 3.04\,\mathrm{A}$, $I_{D,max} = 5.96\,\mathrm{A}$, $\mathrm{PIV} = 13.5\,\mathrm{V}$

Solution 3C.10 **Medium**



Step 1 — Peak output

At the peak of the input the current flows through two diodes in series (one on each side of the bridge), each dropping 0.7 V (tool T1):

$$V_p = \hat{V}_S - 1.4 = 10 - 1.4 = 8.6\,\text{V}, \qquad I_L \approx \frac{V_p}{R_L} = 86\,\text{mA}$$

Step 2 — Filter capacitor

Full-wave: the capacitor is recharged every half period, $T = 1/120\,\text{s}$. Between recharges the capacitor discharges through R_L and loses $V_r \approx V_p \frac{T}{R_L C}$ (tool T4). We need $V_r = 500\,\text{mV}$:

$$C = \frac{V_p T}{R_L V_r} = \frac{8.6 \times 0.008333}{100 \times 0.5} = 1.43\,\text{mF}$$

Step 3 — Diode currents

The diodes only conduct during the short recharge interval, so they must deliver a whole period's charge in a short burst (tool T4):

$$\sqrt{2V_p/V_r} = \sqrt{2 \times 8.6/0.5} = 5.865$$

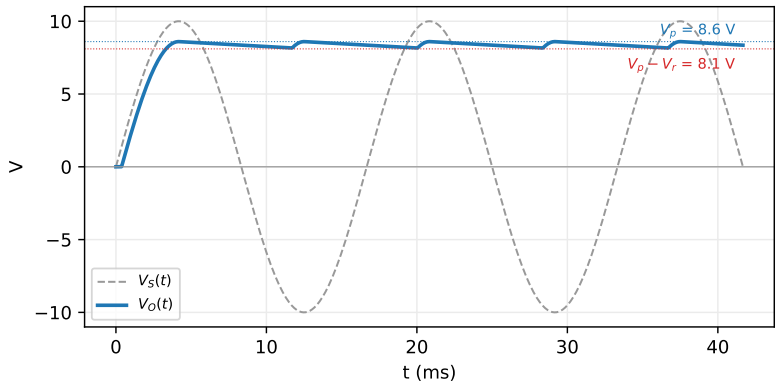
$$I_{D,av} = I_L\left(1 + \pi\sqrt{2V_p/V_r}\right) = 0.086\,(1 + \pi \times 5.865) = 1.67\,\text{A}$$

$$I_{D,max} = I_L\left(1 + 2\pi\sqrt{2V_p/V_r}\right) = 3.26\,\text{A}$$

Step 4 — Peak inverse voltage

When a pair of diodes conducts, each of the other two sees the output plus one diode drop: PIV $= V_p + 0.7 = \hat{V}_S - 0.7 = 9.3\,\text{V}$. The diodes must have a reverse breakdown voltage above the PIV.

Step 5 — Waveforms

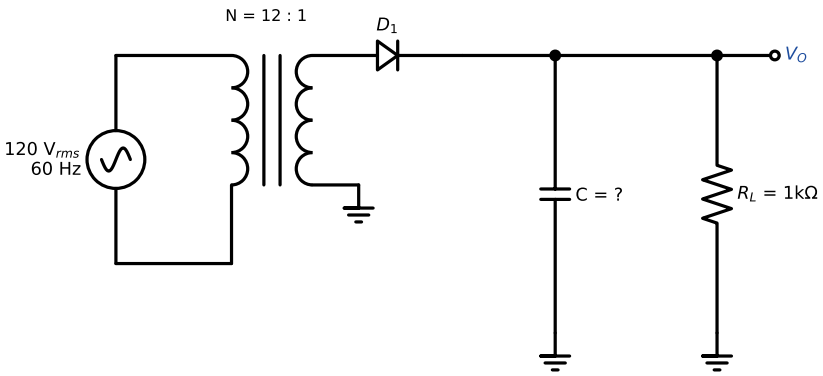


The ripple formula assumes the capacitor discharges for a whole T; in reality it is recharged a little earlier, so the true ripple (plotted) is slightly smaller — the formula errs on the safe side.

Answer

$V_p = 8.6\,\text{V}, \quad C = 1.43\,\text{mF}, \quad I_{D,av} = 1.67\,\text{A}, \quad I_{D,max} = 3.26\,\text{A}, \quad \text{PIV} = 9.3\,\text{V}$

Solution 3C.11 **Medium**



Step 1 — Secondary voltage

The mains peak is $120\sqrt{2} = 170\,\text{V}$, so the secondary peaks at $\frac{170}{N} = 14.2\,\text{V} = \hat{V}_S$.

Step 2 — Peak output

At the peak of the input the current flows through the single diode, each dropping 0.7 V (tool T1):

$$V_p = \hat{V}_S - 0.7 = 14.17 - 0.7 = 13.5\,\text{V}, \qquad I_L \approx \frac{V_p}{R_L} = 13.5\,\text{mA}$$

Step 3 — Filter capacitor

Half-wave: the capacitor is recharged once per period, $T = 1/60\,\text{s}$. Between recharges the capacitor discharges through R_L and loses $V_r \approx V_p \frac{T}{R_L C}$ (tool T4). We need $V_r = V_p - V_{o,min} = 13.47 - 11 = 2.47\,\text{V}$:

$$C = \frac{V_p T}{R_L V_r} = \frac{13.47 \times 0.01667}{1000 \times 2.467} = 91\,\mu\text{F}$$

Step 4 — Diode currents

The diodes only conduct during the short recharge interval, so they must deliver a whole period's charge in a short burst (tool T4):

$$\sqrt{2V_p/V_r} = \sqrt{2 \times 13.47/2.467} = 3.304$$

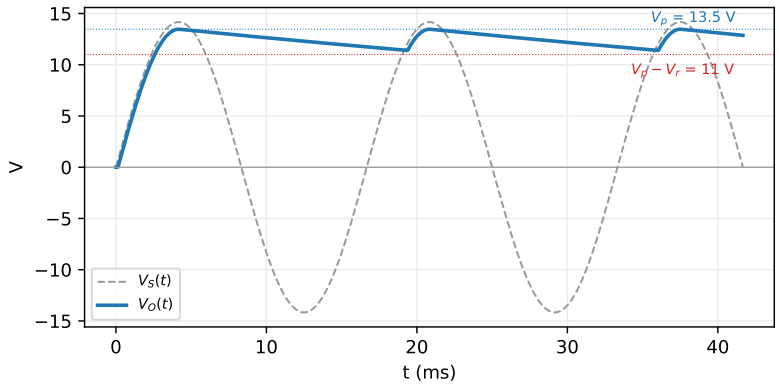
$$I_{D,av} = I_L\left(1 + \pi\sqrt{2V_p/V_r}\right) = 0.01347\,(1 + \pi \times 3.304) = 153\,\text{mA}$$

$$I_{D,max} = I_L\left(1 + 2\pi\sqrt{2V_p/V_r}\right) = 293\,\text{mA}$$

Step 5 — Peak inverse voltage

At the negative peak the anode is at $-\hat{V}_S$ while the capacitor holds the cathode near V_p: PIV $= \hat{V}_S + V_p = 2\hat{V}_S - 0.7 = 27.6\,\text{V}$. The diodes must have a reverse breakdown voltage above the PIV.

Step 6 — Waveforms

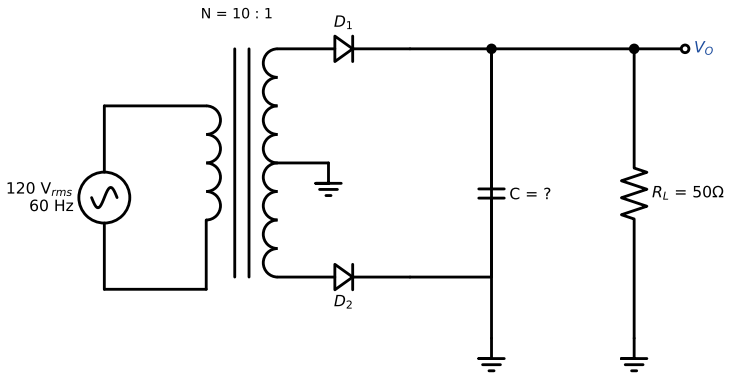


The ripple formula assumes the capacitor discharges for a whole T; in reality it is recharged a little earlier, so the true ripple (plotted) is slightly smaller — the formula errs on the safe side.

Answer

$V_p = 13.5\,\text{V}, \quad C = 91\,\mu\text{F}, \quad I_{D,av} = 153\,\text{mA}, \quad I_{D,max} = 293\,\text{mA}, \quad \text{PIV} = 27.6\,\text{V}$

Solution 3C.12 **Medium**



Step 1 — Secondary voltage

The mains peak is $120\sqrt{2} = 170\,\text{V}$, so the secondary peaks at $\frac{170}{N} = 17\,\text{V}$; each half of the center-tapped secondary gets $\hat{V}_S = 8.5\,\text{V}$.

Step 2 — Peak output

At the peak of the input the current flows through one diode (each half of the secondary has its own diode), each dropping 0.7 V (tool T1):

$$V_p = \hat{V}_S - 0.7 = 8.5 - 0.7 = 7.8\,\text{V}, \qquad I_L \approx \frac{V_p}{R_L} = 156\,\text{mA}$$

Step 3 — Filter capacitor

Full-wave: the capacitor is recharged every half period, $T = 1/120\,\text{s}$. Between recharges the capacitor discharges through R_L and loses $V_r \approx V_p \frac{T}{R_L C}$ (tool T4). We need $V_r = 400\,\text{mV}$:

$$C = \frac{V_p T}{R_L V_r} = \frac{7.8 \times 0.008333}{50 \times 0.4} = 3.25\,\text{mF}$$

Step 4 — Diode currents

The diodes only conduct during the short recharge interval, so they must deliver a whole period's charge in a short burst (tool T4):

$$\sqrt{2V_p/V_r} = \sqrt{2 \times 7.8/0.4} = 6.245$$

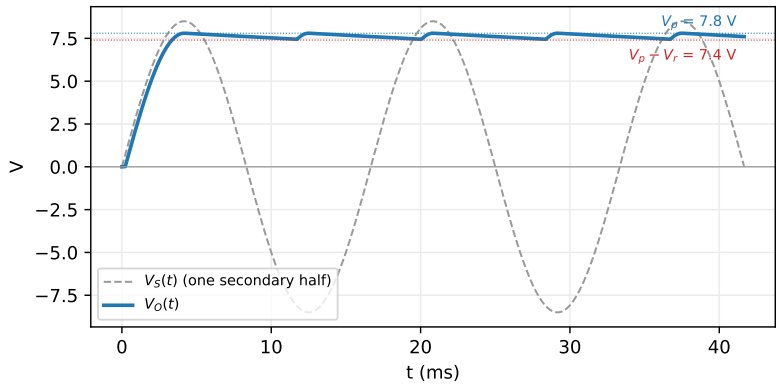
$$I_{D,av} = I_L\left(1 + \pi\sqrt{2V_p/V_r}\right) = 0.156\,(1 + \pi \times 6.245) = 3.22\,\text{A}$$

$$I_{D,max} = I_L\left(1 + 2\pi\sqrt{2V_p/V_r}\right) = 6.28\,\text{A}$$

Step 5 — Peak inverse voltage

When one diode conducts its cathode sits at $V_p = \hat{V}_S - 0.7$ while its own half of the secondary is at $-\hat{V}_S$: $\text{PIV} = 2\hat{V}_S - 0.7 = 16.3\,\text{V}$. The diodes must have a reverse breakdown voltage above the PIV.

Step 6 — Waveforms

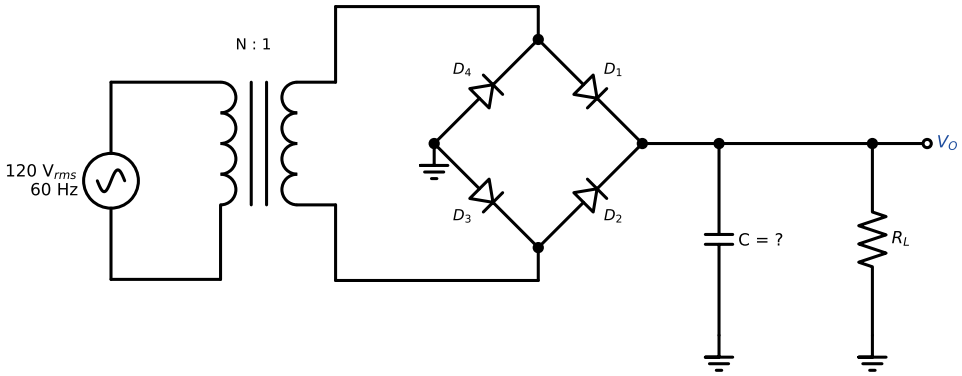


The ripple formula assumes the capacitor discharges for a whole T; in reality it is recharged a little earlier, so the true ripple (plotted) is slightly smaller — the formula errs on the safe side.

Answer

$V_p = 7.8\,\text{V}, \quad C = 3.25\,\text{mF}, \quad I_{D,av} = 3.22\,\text{A}, \quad I_{D,max} = 6.28\,\text{A}, \quad \text{PIV} = 16.3\,\text{V}$

Solution 3C.13 **Hard**



Step 1 — Worst-case load

The ripple is largest at the largest current, so design for $I_L = 400\,\text{mA}$: the load looks like $R_L = \frac{V_o}{I_{L,max}} = \frac{3}{0.4} = 7.5\,\Omega$.

Step 2 — Transformer ratio

The mains peak is $120\sqrt{2} = 170\,\text{V}$. The current passes through two diodes in series (one on each side of the bridge), so the secondary must peak at $\hat{V}_S = V_o + 1.4 = 4.4\,\text{V}$, and $N = \frac{170}{\hat{V}_S} = \frac{170}{4.4} = 38.64$.

Step 3 — Filter capacitor

Full-wave: the capacitor is recharged every half period, $T = 1/120\,\text{s}$. Ripple $V_r = V_p \frac{T}{R_L C}$ and $K_r = \frac{V_r}{V_p} = \frac{T}{R_L C}$ (tool T4), so

$$C = \frac{T}{R_L K_r} = \frac{0.008333}{7.5 \times 0.05} = 22.2\,\text{mF}$$

Step 4 — Diode currents

The diodes only conduct during the short recharge interval, so they must deliver a whole period's charge in a short burst (tool T4):

$$\sqrt{2V_p/V_r} = \sqrt{2 \times 3/0.15} = 6.325$$

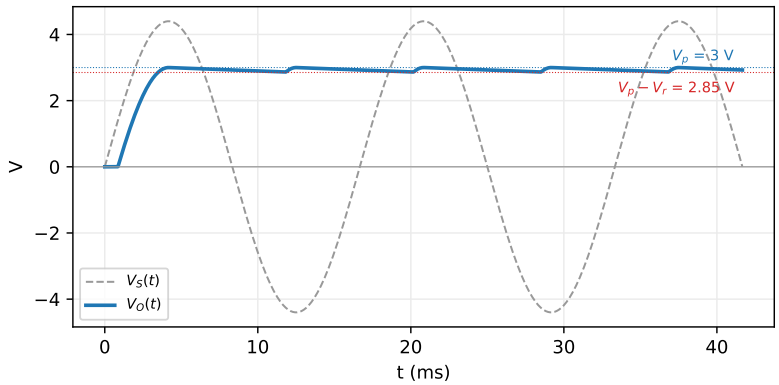
$$I_{D,av} = I_L \left(1 + \pi\sqrt{2V_p/V_r}\right) = 0.4\,(1 + \pi \times 6.325) = 8.35\,\text{A}$$

$$I_{D,max} = I_L \left(1 + 2\pi\sqrt{2V_p/V_r}\right) = 16.3\,\text{A}$$

Step 5 — Peak inverse voltage

When a pair of diodes conducts, each of the other two sees the output plus one diode drop: PIV $= V_p + 0.7 = \hat{V}_S - 0.7 = 3.7\,\text{V}$. The diodes must have a reverse breakdown voltage above the PIV.

Step 6 — Waveforms

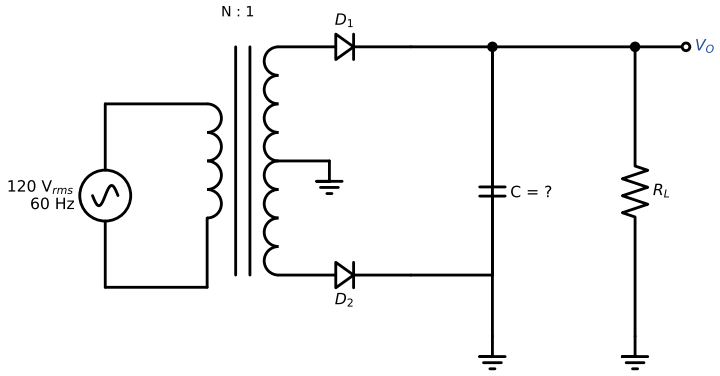


The ripple formula assumes the capacitor discharges for a whole T; in reality it is recharged a little earlier, so the true ripple (plotted) is slightly smaller — the formula errs on the safe side.

Answer

$N = 38.64, \quad C = 22.2\,\text{mF}, \quad R_L = 7.5\,\Omega, \quad I_{D,av} = 8.35\,\text{A}, \quad I_{D,max} = 16.3\,\text{A}, \quad \text{PIV} = 3.7\,\text{V}$

Solution 3C.14 **Hard**



Step 1 — Worst-case load

The ripple is largest at the largest current, so design for $I_L = 1.5\,\text{A}$: the load looks like $R_L = \frac{V_o}{I_{L,max}} = \frac{5}{1.5} = 3.33\,\Omega$.

Step 2 — Transformer ratio

The mains peak is $120\sqrt{2} = 170\,\text{V}$. Each half of the secondary must peak at $\hat{V}_S = V_o + 0.7 = 5.7\,\text{V}$, so the whole secondary peaks at $2\hat{V}_S = 11.4\,\text{V}$ and $N = \frac{170}{2(V_o+0.7)} = \frac{170}{11.4} = 14.91$.

Step 3 — Filter capacitor

Full-wave: the capacitor is recharged every half period, $T = 1/120\,\text{s}$. Ripple $V_r = V_p \frac{T}{R_L C}$ and $K_r = \frac{V_r}{V_p} = \frac{T}{R_L C}$ (tool T4), so

$$C = \frac{T}{R_L K_r} = \frac{0.008333}{3.333 \times 0.1} = 25\,\text{mF}$$

Step 4 — Diode currents

The diodes only conduct during the short recharge interval, so they must deliver a whole period's charge in a short burst (tool T4):

$$\sqrt{2V_p/V_r} = \sqrt{2 \times 5/0.5} = 4.472$$

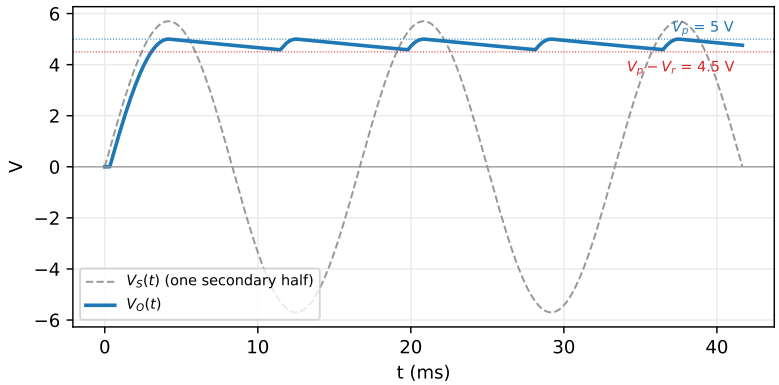
$$I_{D,av} = I_L\left(1 + \pi\sqrt{2V_p/V_r}\right) = 1.5\,(1 + \pi \times 4.472) = 22.6\,\text{A}$$

$$I_{D,max} = I_L\left(1 + 2\pi\sqrt{2V_p/V_r}\right) = 43.6\,\text{A}$$

Step 5 — Peak inverse voltage

When one diode conducts its cathode sits at $V_p = \hat{V}_S - 0.7$ while its own half of the secondary is at $-\hat{V}_S$: $\text{PIV} = 2\hat{V}_S - 0.7 = 10.7\,\text{V}$. The diodes must have a reverse breakdown voltage above the PIV.

Step 6 — Waveforms

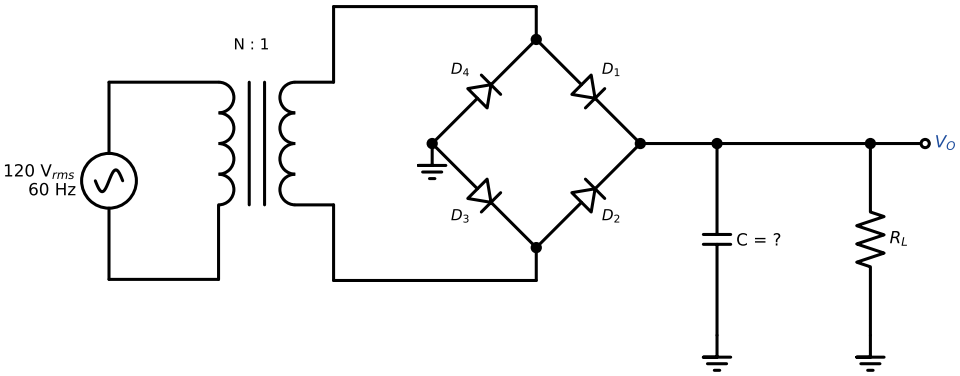


The ripple formula assumes the capacitor discharges for a whole T; in reality it is recharged a little earlier, so the true ripple (plotted) is slightly smaller — the formula errs on the safe side.

Answer

$N = 14.91, \quad C = 25\,\text{mF}, \quad R_L = 3.33\,\Omega, \quad I_{D,av} = 22.6\,\text{A}, \quad I_{D,max} = 43.6\,\text{A}, \quad \text{PIV} = 10.7\,\text{V}$

Solution 3C.15 **Hard**



Step 1 — Worst-case load

The ripple is largest at the largest current, so design for $I_L = 1\,\text{A}$: the load looks like $R_L = \frac{V_o}{I_{L,max}} = \frac{12}{1} = 12\,\Omega$.

Step 2 — Transformer ratio

The mains peak is $120\sqrt{2} = 170\,\text{V}$. The current passes through two diodes in series (one on each side of the bridge), so the secondary must peak at $\hat{V}_S = V_o + 1.4 = 13.4\,\text{V}$, and $N = \frac{170}{\hat{V}_S} = \frac{170}{13.4} = 12.69$.

Step 3 — Filter capacitor

Full-wave: the capacitor is recharged every half period, $T = 1/120\,\text{s}$. Ripple $V_r = V_p \frac{T}{R_L C}$ and $K_r = \frac{V_r}{V_p} = \frac{T}{R_L C}$ (tool T4), so

$$C = \frac{T}{R_L K_r} = \frac{0.008333}{12 \times 0.05} = 13.9\,\text{mF}$$

Step 4 — Diode currents

The diodes only conduct during the short recharge interval, so they must deliver a whole period's charge in a short burst (tool T4):

$$\sqrt{2V_p/V_r} = \sqrt{2 \times 12/0.6} = 6.325$$

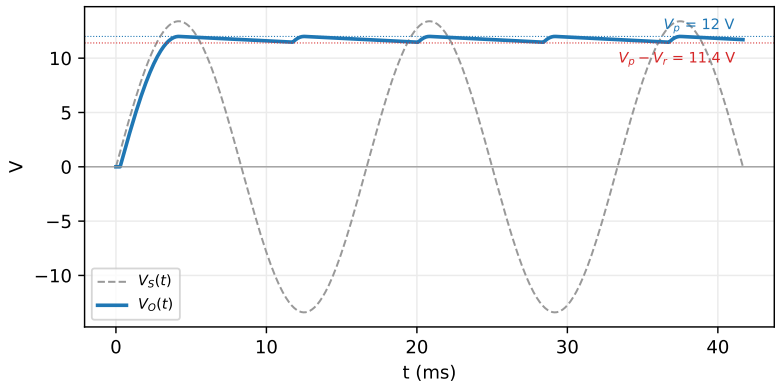
$$I_{D,av} = I_L \left(1 + \pi\sqrt{2V_p/V_r}\right) = 1\,(1 + \pi \times 6.325) = 20.9\,\text{A}$$

$$I_{D,max} = I_L \left(1 + 2\pi\sqrt{2V_p/V_r}\right) = 40.7\,\text{A}$$

Step 5 — Peak inverse voltage

When a pair of diodes conducts, each of the other two sees the output plus one diode drop: PIV $= V_p + 0.7 = \hat{V}_S - 0.7 = 12.7\,\text{V}$. The diodes must have a reverse breakdown voltage above the PIV.

Step 6 — Waveforms

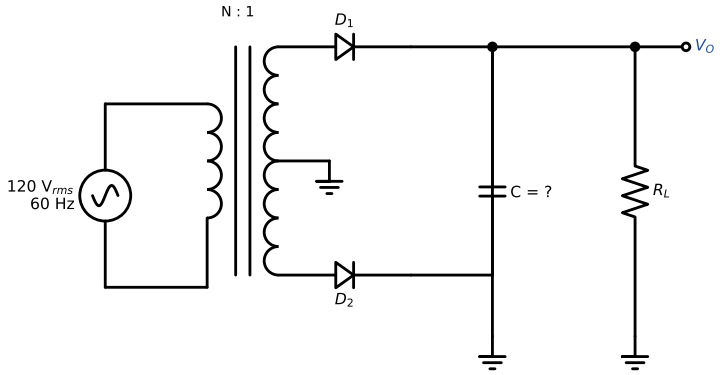


The ripple formula assumes the capacitor discharges for a whole T; in reality it is recharged a little earlier, so the true ripple (plotted) is slightly smaller — the formula errs on the safe side.

Answer

$N = 12.69, \quad C = 13.9\,\text{mF}, \quad R_L = 12\,\Omega, \quad I_{D,av} = 20.9\,\text{A}, \quad I_{D,max} = 40.7\,\text{A}, \quad \text{PIV} = 12.7\,\text{V}$

Solution 3C.16 **Hard**



Step 1 — Worst-case load

The ripple is largest at the largest current, so design for $I_L = 500\,\text{mA}$: the load looks like $R_L = \frac{V_o}{I_{L,max}} = \frac{9}{0.5} = 18\,\Omega$.

Step 2 — Transformer ratio

The mains peak is $120\sqrt{2} = 170\,\text{V}$. Each half of the secondary must peak at $\hat{V}_S = V_o + 0.7 = 9.7\,\text{V}$, so the whole secondary peaks at $2\hat{V}_S = 19.4\,\text{V}$ and $N = \frac{170}{2(V_o+0.7)} = \frac{170}{19.4} = 8.763$.

Step 3 — Filter capacitor

Full-wave: the capacitor is recharged every half period, $T = 1/120\,\text{s}$. Ripple $V_r = V_p \frac{T}{R_L C}$ and $K_r = \frac{V_r}{V_p} = \frac{T}{R_L C}$ (tool T4), so

$$C = \frac{T}{R_L K_r} = \frac{0.008333}{18 \times 0.05} = 9.26\,\text{mF}$$

Step 4 — Diode currents

The diodes only conduct during the short recharge interval, so they must deliver a whole period's charge in a short burst (tool T4):

$$\sqrt{2V_p/V_r} = \sqrt{2 \times 9/0.45} = 6.325$$

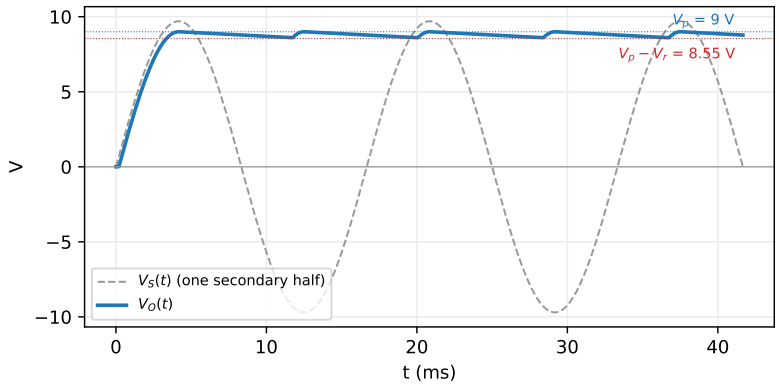
$$I_{D,av} = I_L\left(1 + \pi\sqrt{2V_p/V_r}\right) = 0.5\,(1 + \pi \times 6.325) = 10.4\,\text{A}$$

$$I_{D,max} = I_L\left(1 + 2\pi\sqrt{2V_p/V_r}\right) = 20.4\,\text{A}$$

Step 5 — Peak inverse voltage

When one diode conducts its cathode sits at $V_p = \hat{V}_S - 0.7$ while its own half of the secondary is at $-\hat{V}_S$: $\text{PIV} = 2\hat{V}_S - 0.7 = 18.7\,\text{V}$. The diodes must have a reverse breakdown voltage above the PIV.

Step 6 — Waveforms

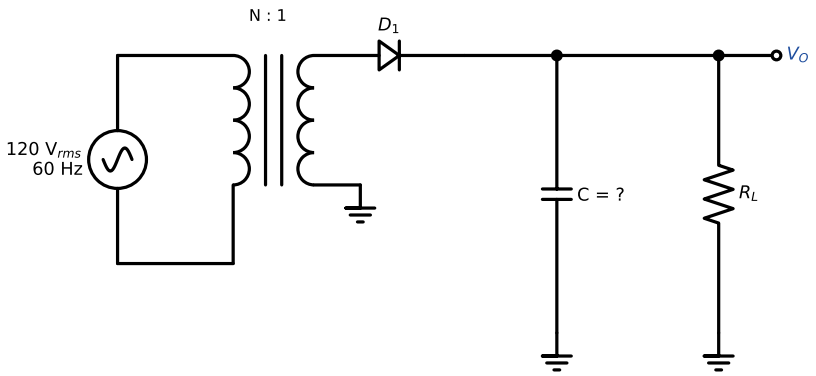


The ripple formula assumes the capacitor discharges for a whole T; in reality it is recharged a little earlier, so the true ripple (plotted) is slightly smaller — the formula errs on the safe side.

Answer

$N = 8.763, \quad C = 9.26\,\text{mF}, \quad R_L = 18\,\Omega, \quad I_{D,av} = 10.4\,\text{A}, \quad I_{D,max} = 20.4\,\text{A}, \quad \text{PIV} = 18.7\,\text{V}$

Solution 3C.17 **Hard**



Step 1 — Worst-case load

The ripple is largest at the largest current, so design for $I_L = 100\,\text{mA}$: the load looks like $R_L = \frac{V_o}{I_{L,max}} = \frac{5}{0.1} = 50\,\Omega$.

Step 2 — Transformer ratio

The mains peak is $120\sqrt{2} = 170\,\text{V}$. The current passes through the single diode, so the secondary must peak at $\hat{V}_S = V_o + 0.7 = 5.7\,\text{V}$, and $N = \frac{170}{\hat{V}_S} = \frac{170}{5.7} = 29.82$.

Step 3 — Filter capacitor

Half-wave: the capacitor is recharged once per period, $T = 1/60\,\text{s}$. Ripple $V_r = V_p \frac{T}{R_L C}$ and $K_r = \frac{V_r}{V_p} = \frac{T}{R_L C}$ (tool T4), so

$$C = \frac{T}{R_L K_r} = \frac{0.01667}{50 \times 0.1} = 3.33\,\text{mF}$$

Step 4 — Diode currents

The diodes only conduct during the short recharge interval, so they must deliver a whole period's charge in a short burst (tool T4):

$$\sqrt{2V_p/V_r} = \sqrt{2 \times 5/0.5} = 4.472$$

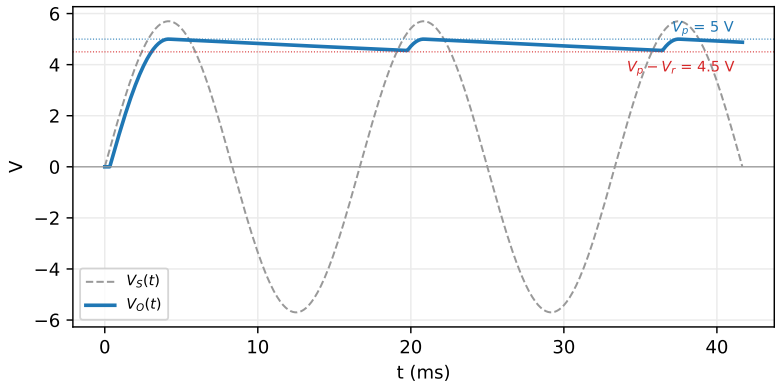
$$I_{D,av} = I_L\left(1 + \pi\sqrt{2V_p/V_r}\right) = 0.1\,(1 + \pi \times 4.472) = 1.5\,\text{A}$$

$$I_{D,max} = I_L\left(1 + 2\pi\sqrt{2V_p/V_r}\right) = 2.91\,\text{A}$$

Step 5 — Peak inverse voltage

At the negative peak the anode is at $-\hat{V}_S$ while the capacitor holds the cathode near V_p: PIV $= \hat{V}_S + V_p = 2\hat{V}_S - 0.7 = 10.7\,\text{V}$. The diodes must have a reverse breakdown voltage above the PIV.

Step 6 — Waveforms

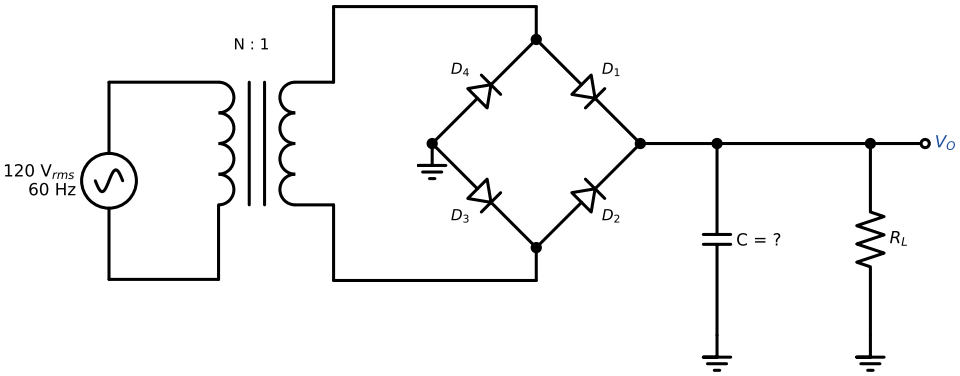


The ripple formula assumes the capacitor discharges for a whole T; in reality it is recharged a little earlier, so the true ripple (plotted) is slightly smaller — the formula errs on the safe side.

Answer

$N = 29.82, \quad C = 3.33\,\text{mF}, \quad R_L = 50\,\Omega, \quad I_{D,av} = 1.5\,\text{A}, \quad I_{D,max} = 2.91\,\text{A}, \quad \text{PIV} = 10.7\,\text{V}$

Solution 3C.18 **Hard**



Step 1 — Worst-case load

The ripple is largest at the largest current, so design for $I_L = 800\,\text{mA}$: the load looks like $R_L = \frac{V_o}{I_{L,max}} = \frac{15}{0.8} = 18.8\,\Omega$.

Step 2 — Transformer ratio

The mains peak is $120\sqrt{2} = 170\,\text{V}$. The current passes through two diodes in series (one on each side of the bridge), so the secondary must peak at $\hat{V}_S = V_o + 1.4 = 16.4\,\text{V}$, and $N = \frac{170}{\hat{V}_S} = \frac{170}{16.4} = 10.37$.

Step 3 — Filter capacitor

Full-wave: the capacitor is recharged every half period, $T = 1/120\,\text{s}$. Ripple $V_r = V_p \frac{T}{R_L C}$ and $K_r = \frac{V_r}{V_p} = \frac{T}{R_L C}$ (tool T4), so

$$C = \frac{T}{R_L K_r} = \frac{0.008333}{18.75 \times 0.05} = 8.89\,\text{mF}$$

Step 4 — Diode currents

The diodes only conduct during the short recharge interval, so they must deliver a whole period's charge in a short burst (tool T4):

$$\sqrt{2V_p/V_r} = \sqrt{2 \times 15/0.75} = 6.325$$

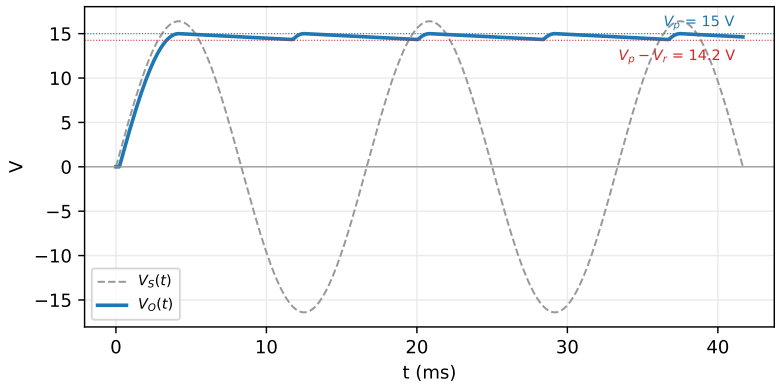
$$I_{D,av} = I_L\left(1 + \pi\sqrt{2V_p/V_r}\right) = 0.8\,(1 + \pi \times 6.325) = 16.7\,\text{A}$$

$$I_{D,max} = I_L\left(1 + 2\pi\sqrt{2V_p/V_r}\right) = 32.6\,\text{A}$$

Step 5 — Peak inverse voltage

When a pair of diodes conducts, each of the other two sees the output plus one diode drop: PIV $= V_p + 0.7 = \hat{V}_S - 0.7 = 15.7\,\text{V}$. The diodes must have a reverse breakdown voltage above the PIV.

Step 6 — Waveforms



The ripple formula assumes the capacitor discharges for a whole T; in reality it is recharged a little earlier, so the true ripple (plotted) is slightly smaller — the formula errs on the safe side.

Answer

$N = 10.37, \quad C = 8.89\,\text{mF}, \quad R_L = 18.8\,\Omega, \quad I_{D,av} = 16.7\,\text{A}, \quad I_{D,max} = 32.6\,\text{A}, \quad \text{PIV} = 15.7\,\text{V}$